PRACTICAL NO. 3

Multi-Dimensional Sales Analytics with Geospatial Visualization

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Course: Data Analytics and Visualization

Course Code: CT-471

Practical Overview

This practical focuses on geospatial and multi-dimensional analytics using Power BI's geographic visuals (Map, Filled Map, Shape Map, Azure Maps) combined with tabular visuals (Table, Matrix), KPI Cards, and Gauge. The star schema includes Time, Geography, and Category dimensions linked to a Sales Fact table.

Visual Type	Questions
Map (Geographic Bubble)	Q1, Q2
Filled Map (Choropleth)	Q3, Q4, Q5
Shape Map (Classification)	Q6, Q7, Q8
Azure Maps (Temporal Heatmap)	Q9, Q10
Table / Matrix	Q11, Q12, Q13, Q14
Card / Multi-row Card	Q15, Q16, Q17, Q18
KPI / Gauge	Q19, Q20

Questions 1–20

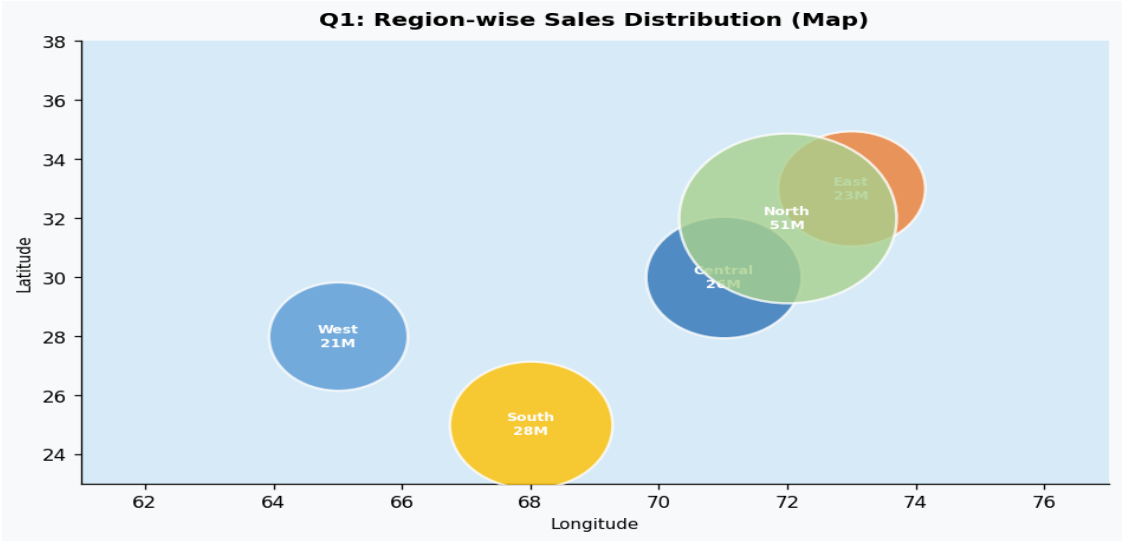
Q1 – Map: Region-wise Sales Distribution

Aggregate total sales by region name and display proportional geographic distribution on a Map visual, with bubble size representing sales magnitude.

SQL Query

```
SELECT REGION_NAME,
       SUM(SALES_AMOUNT) AS TOTAL_SALES
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY = G.GEO_KEY
GROUP  BY REGION_NAME
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Map



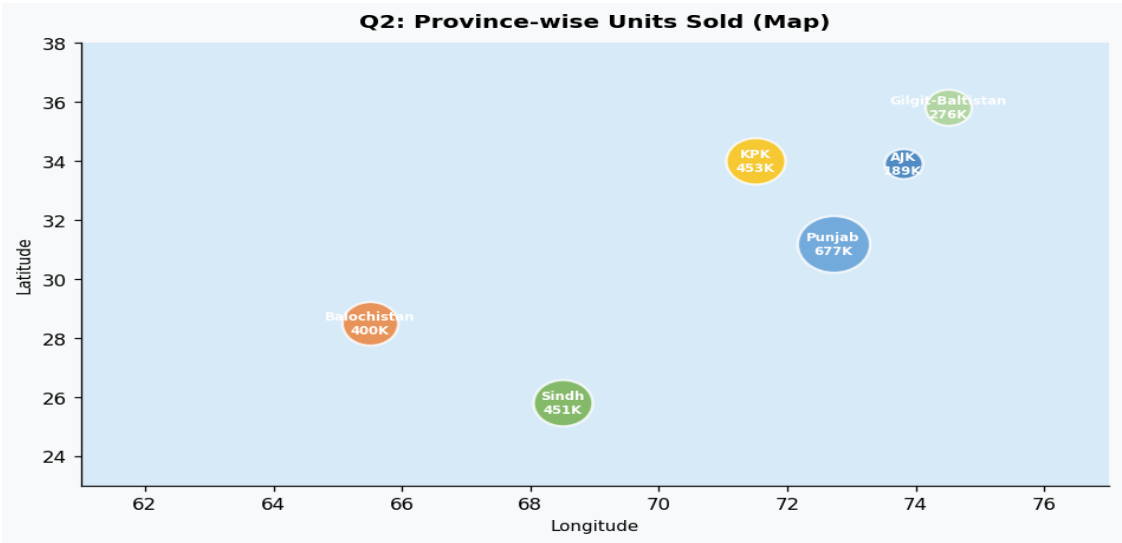
Q2 – Map: Province-wise Units Sold

Compute total units sold per province and visualize spatial volume distribution on a Map, highlighting high-density selling areas.

SQL Query

```
SELECT STATE_PROVINCE_NAME,
       SUM(UNITS_SOLD) AS TOTAL_UNITS
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY = G.GEO_KEY
GROUP  BY STATE_PROVINCE_NAME
ORDER  BY TOTAL_UNITS DESC;
```

Power BI Visual: Map



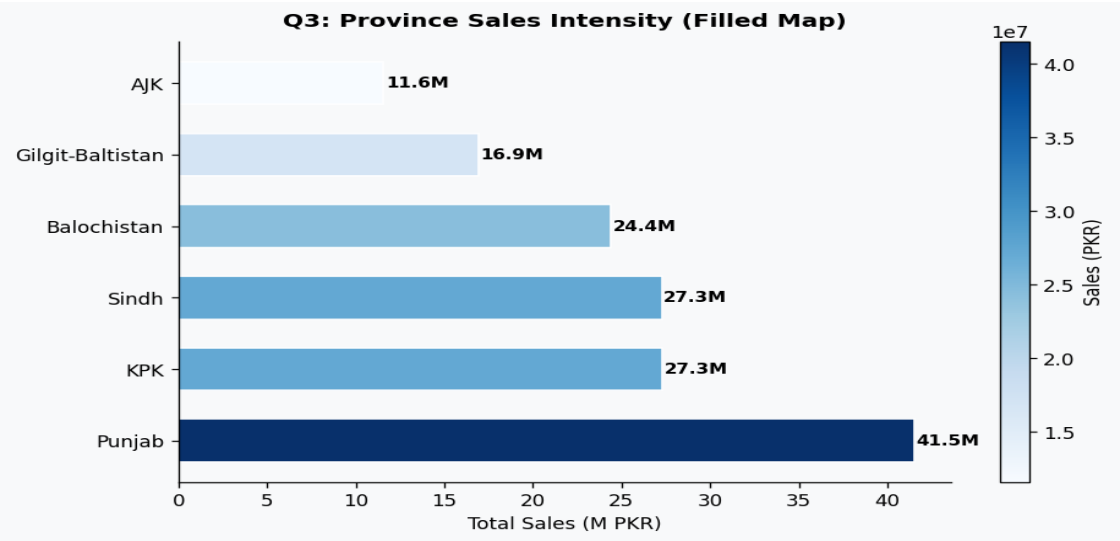
Q3 – Filled Map: Province Sales Intensity

Calculate total sales per province and render color intensity on a Filled Map — darker shades indicate higher sales density.

SQL Query

```
SELECT STATE_PROVINCE_NAME,
       SUM(SALES_AMOUNT) AS TOTAL_SALES
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY = G.GEO_KEY
GROUP  BY STATE_PROVINCE_NAME
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Filled Map



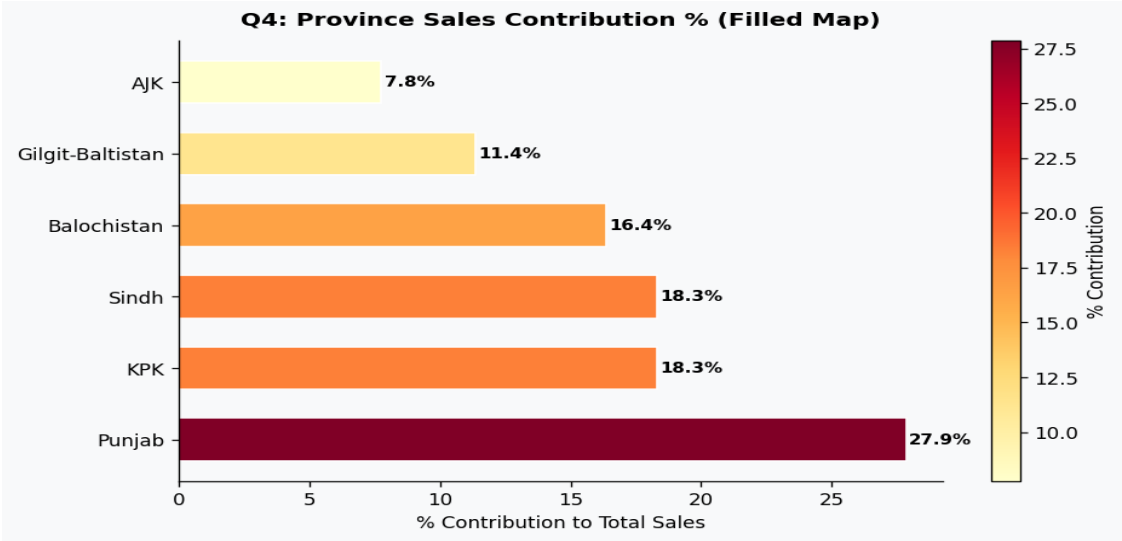
Q4 – Filled Map: Province Contribution Percentage

Compute each province's percentage contribution to national sales and display relative importance as shading intensity on a Filled Map.

SQL Query

```
SELECT STATE_PROVINCE_NAME,
       SUM(SALES_AMOUNT)                                    AS TOTAL_SALES,
       ROUND(SUM(SALES_AMOUNT)*100.0 /
             SUM(SUM(SALES_AMOUNT)) OVER (), 2)             AS PCT_CONTRIBUTION
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY = G.GEO_KEY
GROUP  BY STATE_PROVINCE_NAME
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Filled Map



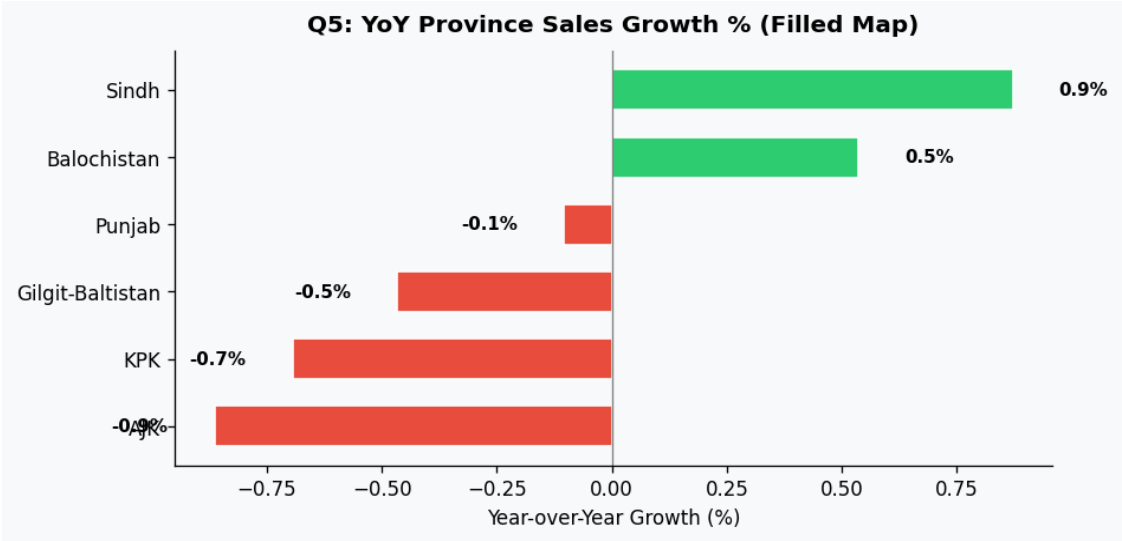
Q5 – Filled Map: Year-over-Year Province Growth

Calculate year-over-year province growth and visualize on a Filled Map — green shading for positive growth, red for decline.

SQL Query

```
WITH YearSales AS (
  SELECT G.STATE_PROVINCE_NAME, T.YEAR_NAME,
         SUM(F.SALES_AMOUNT) AS TOTAL_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.STATE_PROVINCE_NAME, T.YEAR_NAME
)
SELECT STATE_PROVINCE_NAME,
       ROUND((MAX(CASE WHEN YEAR_NAME=2024 THEN TOTAL_SALES END) -
              MAX(CASE WHEN YEAR_NAME=2023 THEN TOTAL_SALES END))*100.0 /
             NULLIF(MAX(CASE WHEN YEAR_NAME=2023 THEN TOTAL_SALES END),0), 2) AS
YOY_GROWTH
FROM   YearSales
GROUP  BY STATE_PROVINCE_NAME;
```

Power BI Visual: Filled Map



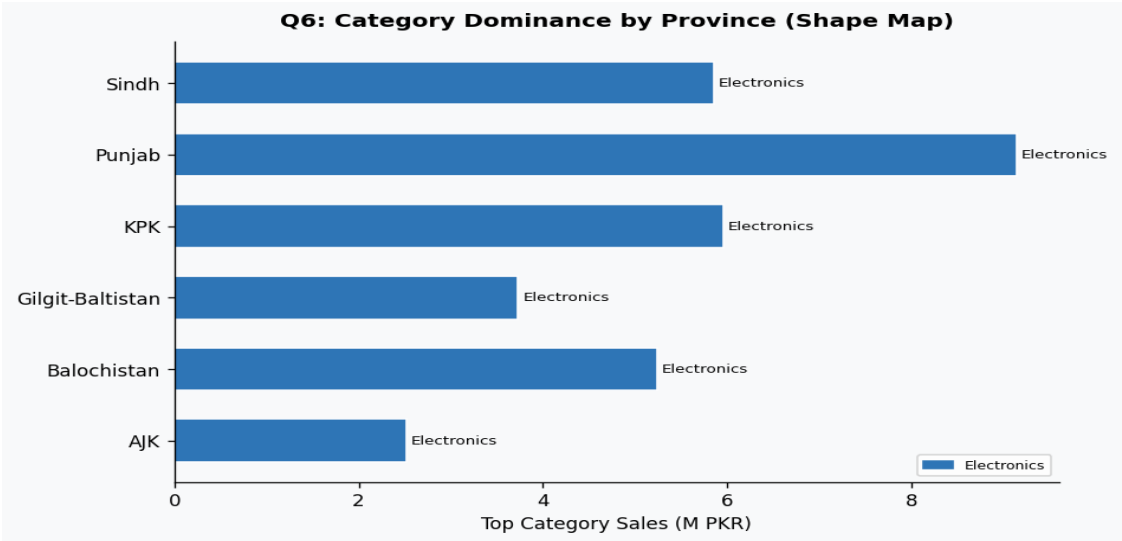
Q6 – Shape Map: Category Dominance by Province

Identify the top-selling category in each province using window functions and visualize category dominance on a Shape Map.

SQL Query

```
SELECT G.STATE_PROVINCE_NAME, C.CATEGORY_NAME,
       SUM(F.SALES_AMOUNT) AS TOTAL_SALES
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
GROUP  BY G.STATE_PROVINCE_NAME, C.CATEGORY_NAME
QUALIFY RANK() OVER (PARTITION BY STATE_PROVINCE_NAME
                     ORDER BY SUM(SALES_AMOUNT) DESC) = 1;
```

Power BI Visual: Shape Map



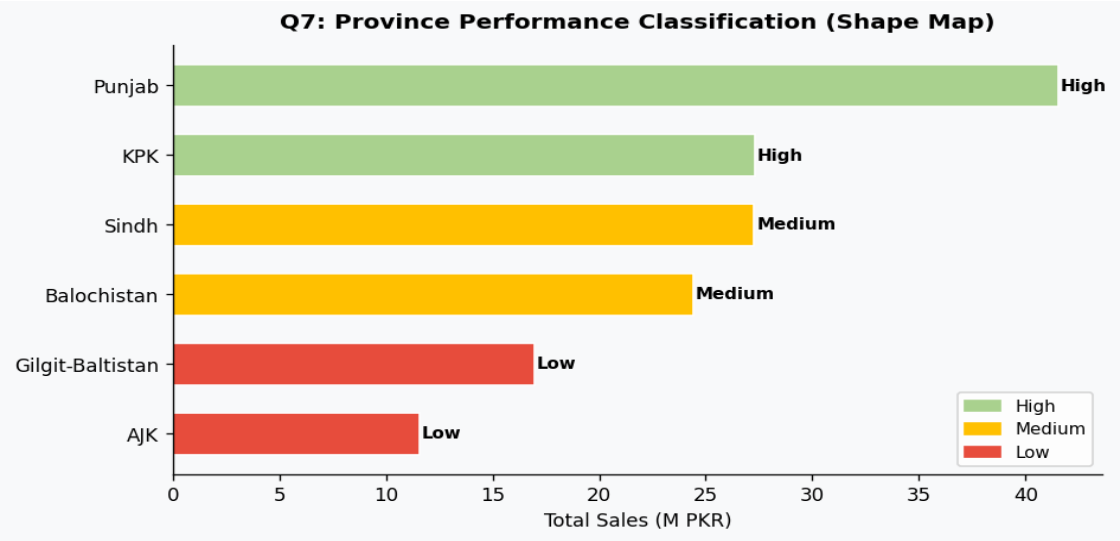
Q7 – Shape Map: Province Performance Classification

Classify provinces into High, Medium, Low groups using PERCENTILE logic and visualize tiered performance on a Shape Map.

SQL Query

```
WITH ProvinceSales AS (
  SELECT G.STATE_PROVINCE_NAME, SUM(F.SALES_AMOUNT) AS TOTAL_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  GROUP  BY G.STATE_PROVINCE_NAME
)
SELECT STATE_PROVINCE_NAME, TOTAL_SALES,
       CASE WHEN TOTAL_SALES >= PERCENTILE_CONT(0.67)
                                WITHIN GROUP (ORDER BY TOTAL_SALES) OVER ()
            THEN 'High'
            WHEN TOTAL_SALES >= PERCENTILE_CONT(0.33)
                                WITHIN GROUP (ORDER BY TOTAL_SALES) OVER ()
            THEN 'Medium'
            ELSE 'Low' END AS PERFORMANCE_CLASS
FROM   ProvinceSales;
```

Power BI Visual: Shape Map



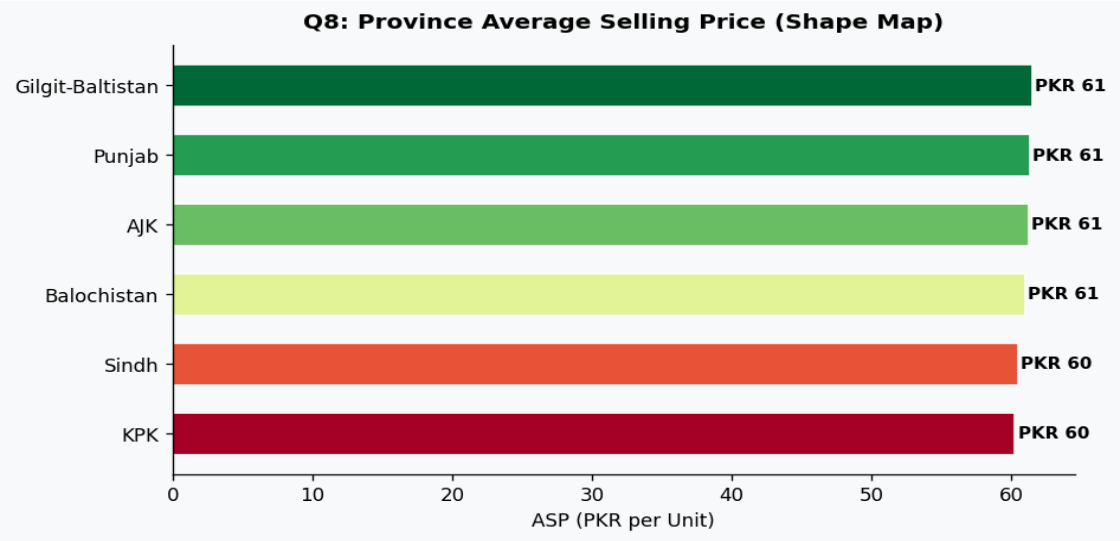
Q8 – Shape Map: Province Average Selling Price

Compute average selling price (sales ÷ units) per province and visualize pricing behavior using gradient shading on a Shape Map.

SQL Query

```
SELECT G.STATE_PROVINCE_NAME,
       ROUND(SUM(F.SALES_AMOUNT) /
             NULLIF(SUM(F.UNITS_SOLD),0), 2)  AS AVG_SELLING_PRICE
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
GROUP  BY G.STATE_PROVINCE_NAME
ORDER  BY AVG_SELLING_PRICE DESC;
```

Power BI Visual: Shape Map



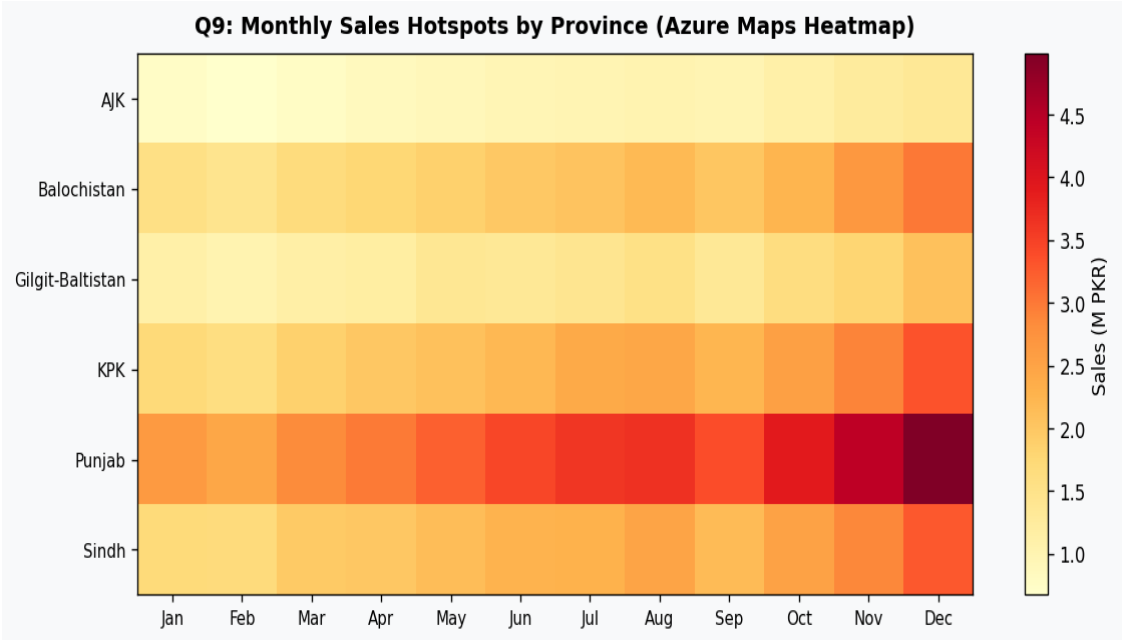
Q9 – Azure Maps: Monthly Sales Hotspots

Calculate monthly sales by province across both years to generate time-series hotspot data for Azure Maps temporal analysis.

SQL Query

```
SELECT G.STATE_PROVINCE_NAME, T.YEAR_NAME, T.MONTH_LONG_NAME,
       SUM(F.SALES_AMOUNT) AS MONTHLY_SALES
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
GROUP  BY G.STATE_PROVINCE_NAME, T.YEAR_NAME, T.MONTH_LONG_NAME
ORDER  BY T.YEAR_NAME, T.MONTH_NUMBER;
```

Power BI Visual: Azure Maps



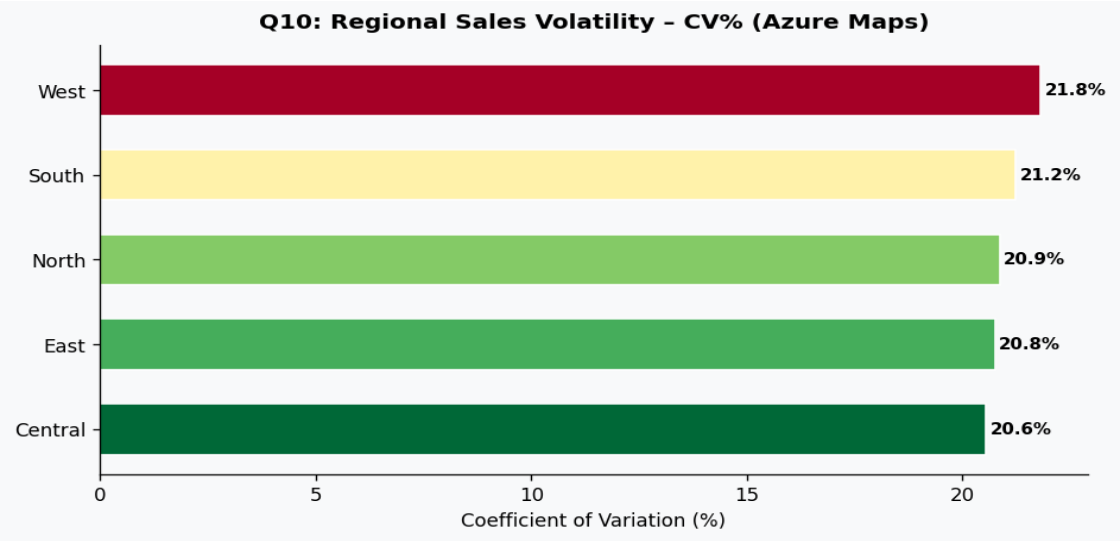
Q10 – Azure Maps: Regional Sales Volatility

Compute each region's monthly sales variability using STDDEV window function; visualize stable vs volatile regions on Azure Maps.

SQL Query

```
SELECT G.REGION_NAME, T.YEAR_NAME, T.MONTH_LONG_NAME,
       SUM(F.SALES_AMOUNT)     AS MONTHLY_SALES,
       STDDEV(SUM(F.SALES_AMOUNT))
             OVER (PARTITION BY G.REGION_NAME)  AS SALES_VOLATILITY
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
GROUP  BY G.REGION_NAME, T.YEAR_NAME, T.MONTH_LONG_NAME;
```

Power BI Visual: Azure Maps



Q11 – Table: Detailed Sales Summary

Detailed summary table showing Year, Month, Region, Category, Total Sales and Units — the core reporting dataset.

SQL Query

```
SELECT T.YEAR_NAME, T.MONTH_LONG_NAME, G.REGION_NAME,
       C.CATEGORY_NAME,
       SUM(F.SALES_AMOUNT) AS TOTAL_SALES,
       SUM(F.UNITS_SOLD)   AS TOTAL_UNITS
FROM   SALES_FACT F
JOIN   TIME_DIM T     ON F.TIME_KEY=T.TIME_KEY
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C  ON F.CAT_KEY=C.CAT_KEY
GROUP  BY T.YEAR_NAME, T.MONTH_LONG_NAME, G.REGION_NAME, C.CATEGORY_NAME
ORDER  BY T.YEAR_NAME, T.MONTH_NUMBER, G.REGION_NAME;
```

Power BI Visual: Table

Q11: Detailed Sales Summary Table (2024 Sample)

YEAR	Month	REGION	CATEGORY	Total Sales	Total Units
2024	Jan	Central	Books	PKR 67.1K	1,987
2024	Jan	Central	Clothing	PKR 140.2K	2,338
2024	Jan	Central	Electronics	PKR 180.7K	740
2024	Jan	Central	Food & Beverages	PKR 128.7K	4,828
2024	Jan	Central	Furniture	PKR 96.7K	291
2024	Jan	Central	Sports	PKR 105.0K	1,089
2024	Jan	Central	Toys	PKR 81.5K	1,632
2024	Jan	East	Books	PKR 68.6K	2,058
2024	Jan	East	Clothing	PKR 127.1K	2,112
2024	Jan	East	Electronics	PKR 141.0K	555

Q12 – Table: Category Consistency Analysis

Calculate average monthly sales and variance per category to assess consistency. Low variance = consistent performer.

SQL Query

```
SELECT C.CATEGORY_NAME,
       ROUND(AVG(MONTHLY_SALES), 2)  AS AVG_MONTHLY_SALES,
       ROUND(VAR_SAMP(MONTHLY_SALES), 2) AS SALES_VARIANCE
FROM (
  SELECT C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_LONG_NAME,
         SUM(F.SALES_AMOUNT) AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  GROUP  BY C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_LONG_NAME
) t
GROUP  BY C.CATEGORY_NAME
ORDER  BY AVG_MONTHLY_SALES DESC;
```

Power BI Visual: Table

Q12: Category Consistency Analysis (Table)

Category	Avg Monthly Sales	Variance
Books	PKR 537.1K	12.99B
Clothing	PKR 1076.4K	48.47B
Electronics	PKR 1352.4K	84.22B
Food & Beverages	PKR 993.5K	45.48B
Furniture	PKR 806.0K	27.51B
Sports	PKR 770.0K	25.87B
Toys	PKR 675.1K	20.20B

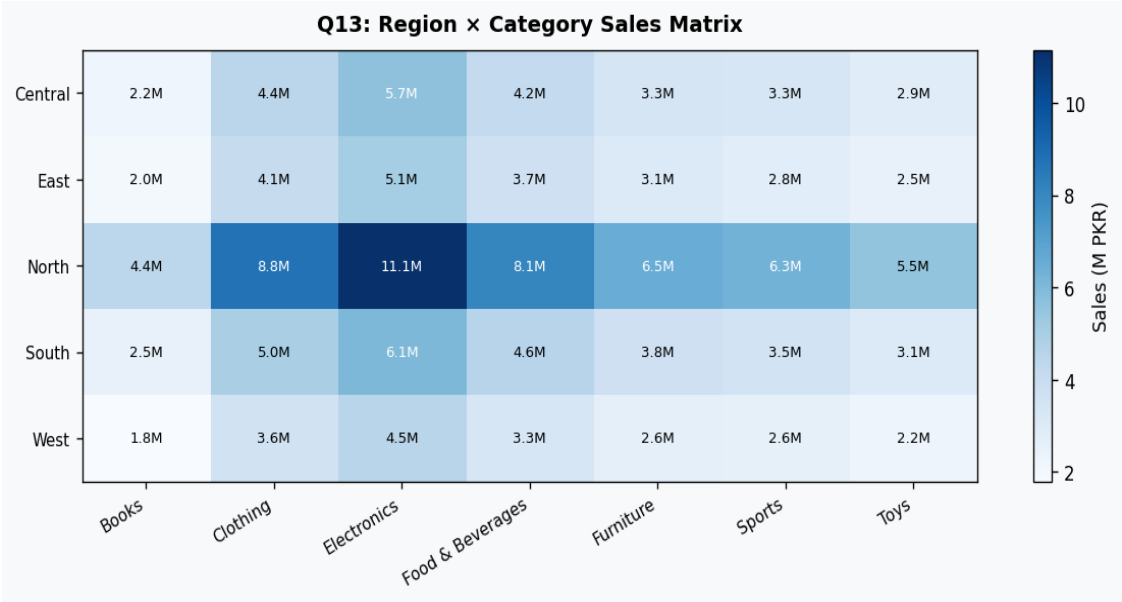
Q13 – Matrix: Region x Category Sales Matrix

Cross-dimensional matrix of Region × Category sales — identifies which region-category combinations drive most revenue.

SQL Query

```
SELECT G.REGION_NAME, C.CATEGORY_NAME,
       SUM(F.SALES_AMOUNT) AS TOTAL_SALES
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C  ON F.CAT_KEY=C.CAT_KEY
GROUP  BY G.REGION_NAME, C.CATEGORY_NAME
ORDER  BY G.REGION_NAME, TOTAL_SALES DESC;
```

Power BI Visual: Matrix



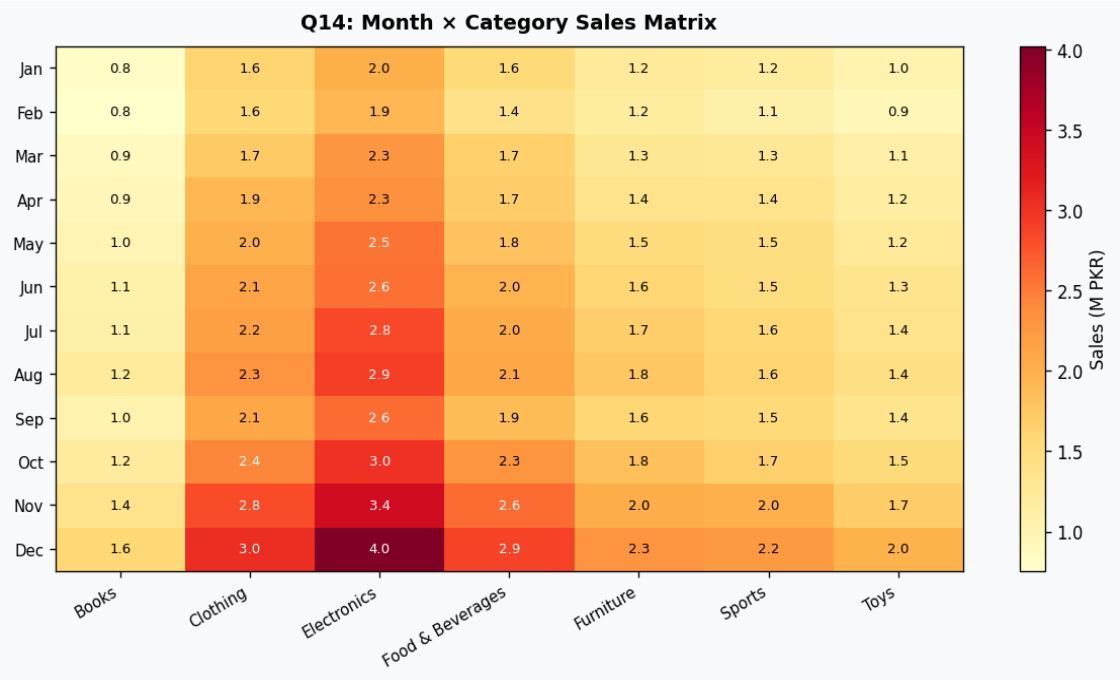
Q14 – Matrix: Month x Category Trend

Monthly × Category sales matrix for temporal trend analysis across the 12-month cycle.

SQL Query

```
SELECT T.MONTH_LONG_NAME, C.CATEGORY_NAME,
       SUM(F.SALES_AMOUNT) AS TOTAL_SALES
FROM   SALES_FACT F
JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
GROUP  BY T.MONTH_LONG_NAME, T.MONTH_NUMBER, C.CATEGORY_NAME
ORDER  BY T.MONTH_NUMBER, C.CATEGORY_NAME;
```

Power BI Visual: Matrix



Q15 – Card: Total Sales Indicator

Single-value KPI Card showing total company-wide sales across all years, regions, and categories.

SQL Query

```
SELECT SUM(SALES_AMOUNT) AS TOTAL_SALES
FROM   SALES_FACT;
```

Power BI Visual: Card

Q16 – Card: Total Units Indicator

Single-value KPI Card showing total units sold — the volume counterpart to the Total Sales Card.

SQL Query

```
SELECT SUM(UNITS_SOLD) AS TOTAL_UNITS
FROM   SALES_FACT;
```

Power BI Visual: Card

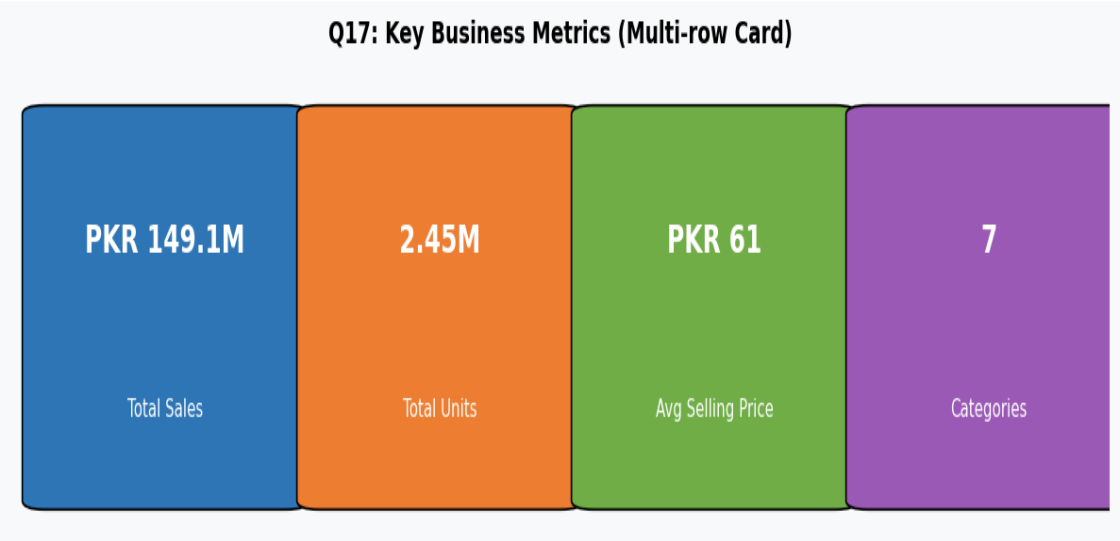
Q17 – Multi-row Card: Key Business Metrics

Multi-row card displaying four key metrics: Total Sales, Total Units, Average Selling Price, and Category Count.

SQL Query

```
SELECT SUM(F.SALES_AMOUNT)                          AS TOTAL_SALES,
       SUM(F.UNITS_SOLD)                            AS TOTAL_UNITS,
       ROUND(SUM(F.SALES_AMOUNT)/
             NULLIF(SUM(F.UNITS_SOLD),0), 2)        AS AVG_SELLING_PRICE,
       COUNT(DISTINCT C.CATEGORY_NAME)              AS DISTINCT_CATEGORIES
FROM   SALES_FACT F
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY;
```

Power BI Visual: Multi-row Card



Q18 – Multi-row Card: Regional Snapshot

Region-level multi-row card with sales, units, and contribution percentage for a quick regional snapshot.

SQL Query

```
SELECT G.REGION_NAME,
       SUM(F.SALES_AMOUNT)                                        AS TOTAL_SALES,
       SUM(F.UNITS_SOLD)                                         AS TOTAL_UNITS,
       ROUND(SUM(F.SALES_AMOUNT)*100.0/SUM(SUM(F.SALES_AMOUNT)) OVER (), 2) AS
CONTRIBUTION_PCT
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
GROUP  BY G.REGION_NAME
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Multi-row Card

Q18: Regional Snapshot (Multi-row Card)

Region	Total Sales	Total Units	Contribution %
Central	PKR 26.0M	428K	17.4%
East	PKR 23.3M	379K	15.6%
North	PKR 50.7M	831K	34.0%
South	PKR 28.4M	467K	19.1%
West	PKR 20.7M	342K	13.9%

Q19 – KPI: Month-over-Month Sales Growth

KPI visual showing current month sales, previous month sales, and month-over-month growth rate with trend arrow.

SQL Query

```
WITH MonthlyData AS (
  SELECT T.YEAR_NAME, T.MONTH_NUMBER, T.MONTH_LONG_NAME,
         SUM(F.SALES_AMOUNT) AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY T.YEAR_NAME, T.MONTH_NUMBER, T.MONTH_LONG_NAME
)
```

```
SELECT YEAR_NAME, MONTH_LONG_NAME, MONTHLY_SALES,
       LAG(MONTHLY_SALES) OVER (ORDER BY YEAR_NAME, MONTH_NUMBER)  AS PREV_SALES,
       ROUND((MONTHLY_SALES - LAG(MONTHLY_SALES)
              OVER (ORDER BY YEAR_NAME, MONTH_NUMBER))*100.0 /
             NULLIF(LAG(MONTHLY_SALES)
              OVER (ORDER BY YEAR_NAME, MONTH_NUMBER), 0), 2)      AS GROWTH_RATE
FROM   MonthlyData
ORDER  BY YEAR_NAME, MONTH_NUMBER;
```

Power BI Visual: KPI

Q20 – Gauge: Sales vs Historical Benchmark

Gauge visual comparing current month sales against the historical monthly average benchmark.

SQL Query

```
WITH MonthlyStats AS (
  SELECT T.YEAR_NAME, T.MONTH_NUMBER, SUM(F.SALES_AMOUNT) AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY T.YEAR_NAME, T.MONTH_NUMBER
)
SELECT YEAR_NAME, MONTH_NUMBER,
       MONTHLY_SALES                              AS CURRENT_SALES,
       ROUND(AVG(MONTHLY_SALES) OVER (), 2)       AS HISTORICAL_BENCHMARK,
       ROUND((MONTHLY_SALES - AVG(MONTHLY_SALES) OVER ())*100.0 /
             AVG(MONTHLY_SALES) OVER (), 2)       AS VS_BENCHMARK_PCT
FROM   MonthlyStats
ORDER  BY YEAR_NAME, MONTH_NUMBER;
```

Power BI Visual: Gauge

Q20: Sales vs Historical Benchmark (Gauge)
PKR 9.06M
Benchmark: 6.21M
ABOVE TARGET